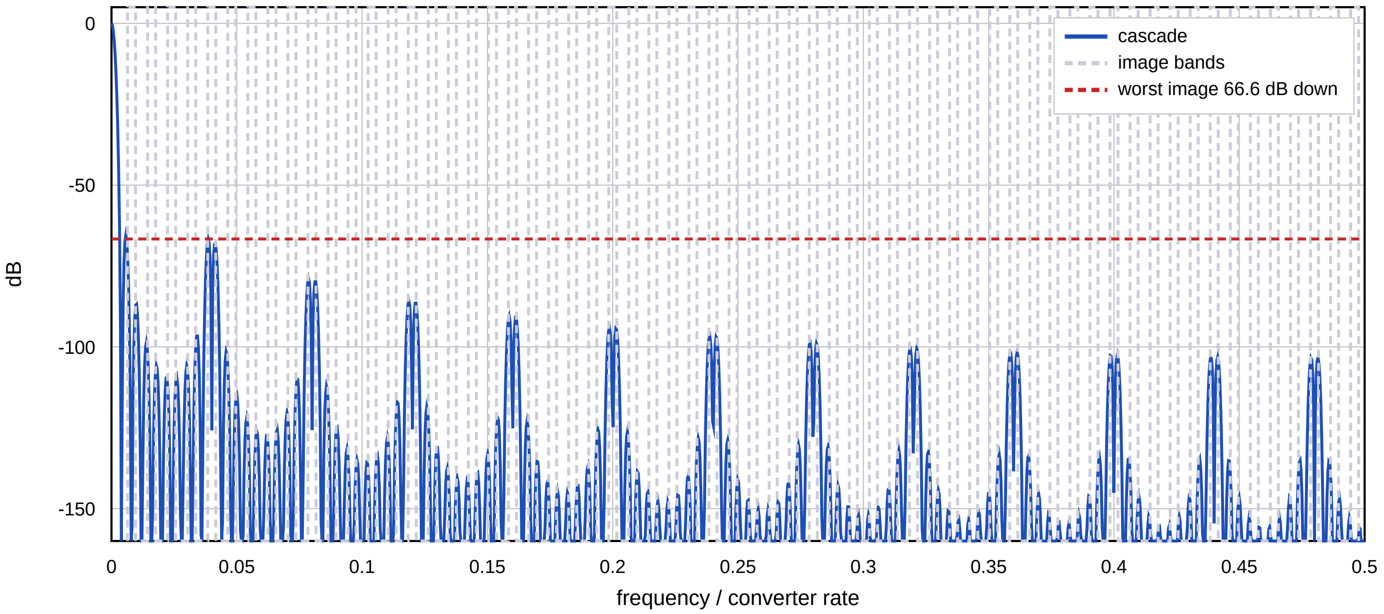
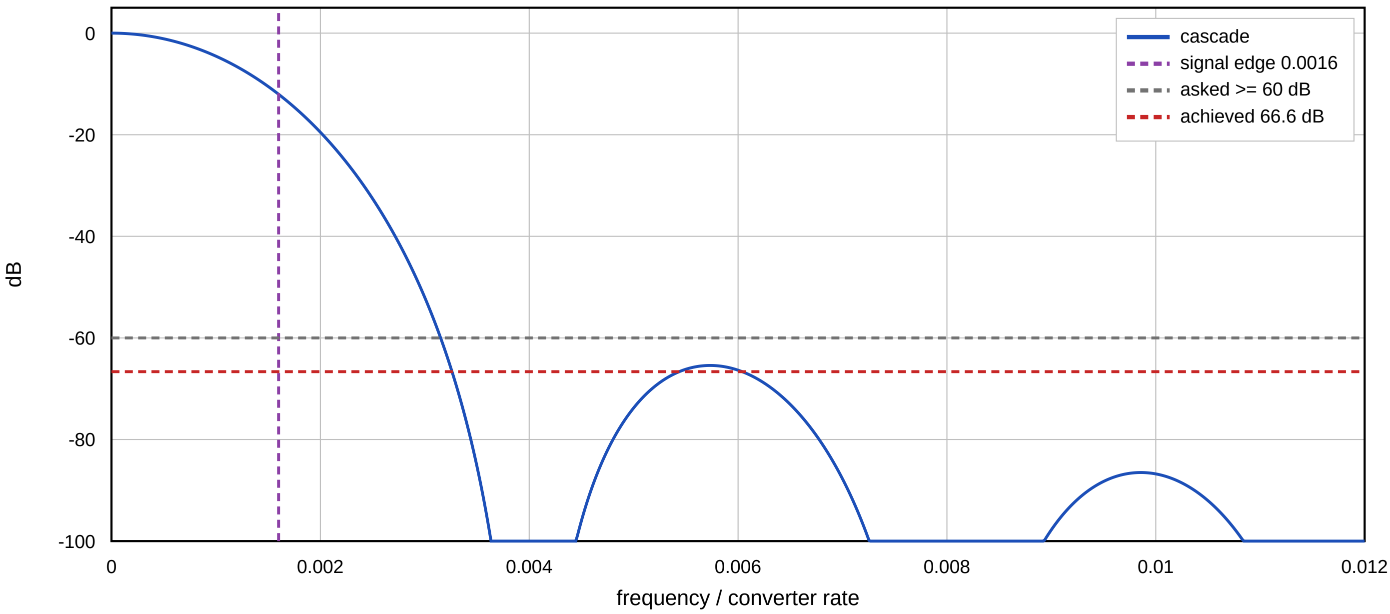
Interpolation Chain - Derived Design

1000.0000 kHz x 125 = 125.000 MHz, split [5, 25], 200.000 kHz signal

Composite magnitude across the converter band



The signal and its first image



Stages

stage 1: x5 N=5 M=2 combs at 1.0000 MHz, integrators at 5.000 MHz
  in 16 bits, uniform accumulator 31 bits, tapered widths [17, 18, 19, 20, 21, 21, 24, 26, 28, 31]
  465 register bits uniform, 320 as built (lossless), bound 320
  widths as built [17, 18, 19, 20, 21, 21, 24, 26, 28, 31]

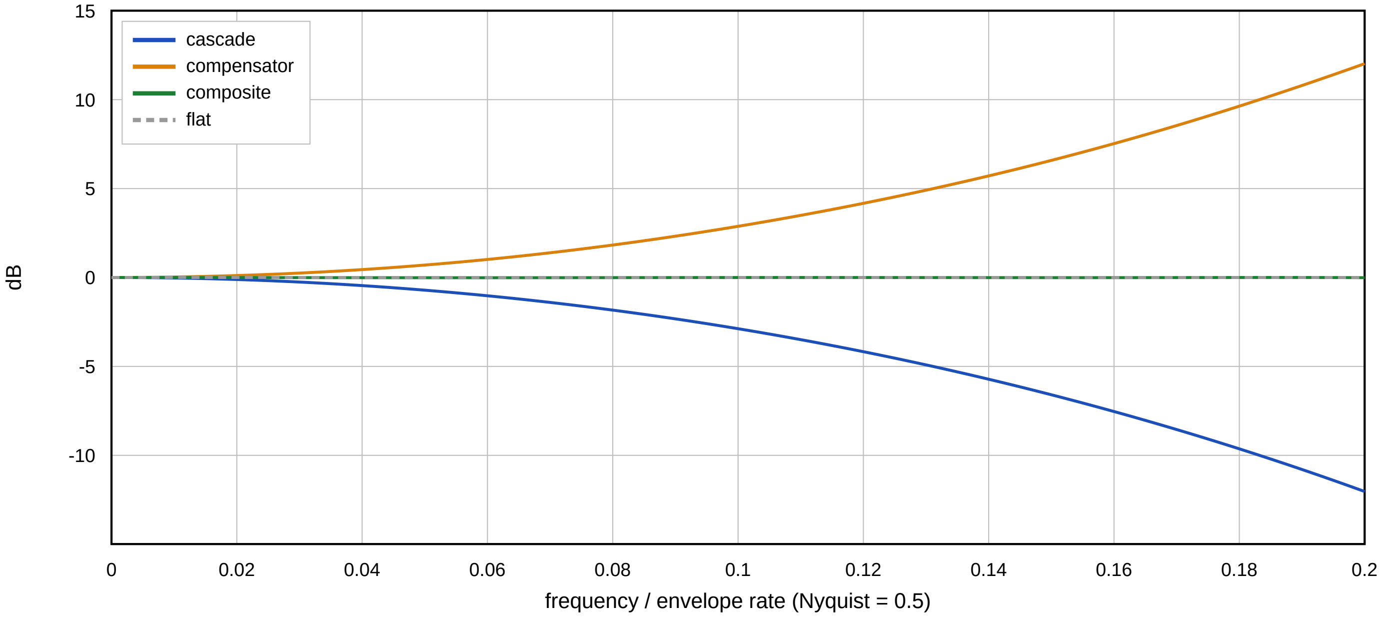
stage 2: x25 N=2 M=1 combs at 5.0000 MHz, integrators at 125.000 MHz
  in 31 bits, uniform accumulator 36 bits, tapered widths [32, 33, 32, 36]
  144 register bits uniform, 134 as built (lossless), bound 133
  widths as built [32, 33, 33, 36]

The combs run at the stage's input rate and the integrators at its output rate, so an integrator bit costs R times what a comb bit costs. Depth belongs early and rate late. The tapered widths are lossless: each stage sized to its own growth bound holds its value exactly, so a tapered interpolator is bit-identical to a uniform one.

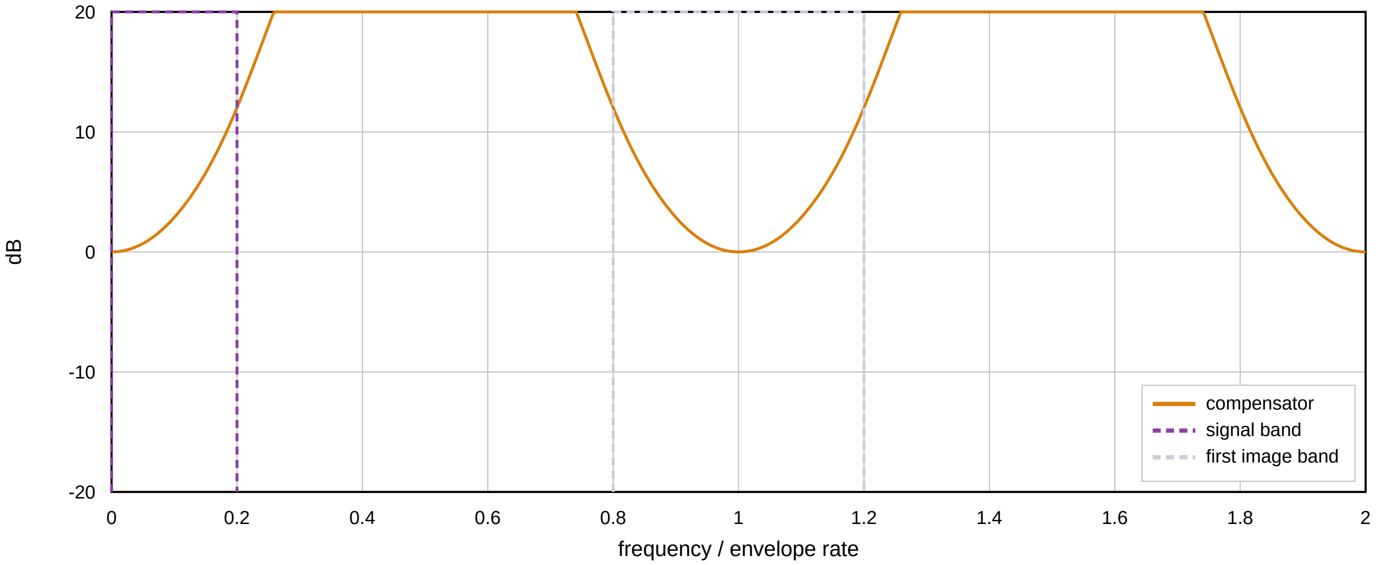
Compensation and Result

9 taps, 16-bit coefficients, at the 1.0000 MHz envelope rate

Droop, pre-compensator, and the two together - the signal band



The compensator repeats, so it lifts every image as much as the signal



Figures

images .... 66.6 dB down (asked >= 60.0)
   worst image at .... 4.8297 MHz
passband ripple ...... 0.0147 dB (asked <= 0.100)
uncompensated droop .. -11.986 dB at the band edge
converter floor ...... 86.0 dB at 14 bits (the chain adds none)
register bits .... 609 uniform, 454 as built, 453 at the exact bound
compensator DC gain .. 1.0000, shift 10
compensator headroom . 23 bits for any input, 18 for in-band only
   norms .... l1 81.1289, passband peak 3.9907
adder depth .... 5 deep in the combs, 1 in the integrators
runner-up .... split [5, 25] N=[5, 2] M=[2, 2], 693 register bits

taps .... [304, -2290, 8288, -18223, 24866, -18223, 8288, -2290, 304]

Build to the any-input headroom unless the envelope is band-limited.

The in-band figure is the peak gain an in-band sinusoid sees. The any-input figure is the sum of the taps' magnitudes, which is the bound for an arbitrary bounded input -- and a transmit envelope is not band-limited: switch-on, a burst boundary and a modulation change are all steps, and a step is exactly the input that reaches it. Choosing the narrower figure for a burst transmitter is how a compensator saturates on the first sample.

More taps will not improve image rejection.

The compensator runs before the rate change, at the envelope rate, so its response is periodic and the image at k±u sees exactly the gain the signal at u sees -- the plot above